



Contact



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The Hong Kong University of Science
and Technology
Clear Water Bay, Kowloon
Hong Kong



[Nicolas Patricio](#)

Education

MPhil in Civil and Environmental Engineering

(Expected Feb 2028)

The Hong Kong University of
Science and Technology

B.S in Agricultural Engineering (Dec 2025)

University of São Paulo, Brazil
GPA: 3.0/4

Skills & Expertise

- CEA (Controlled Environment Agriculture)
- Hydroponic Systems Design
- Pathogen Management in Closed Systems

Curriculum Vitae

Nicolas Paim Patricio

Graduate student in Civil and Environmental Engineering with an interest in nanobubble technology for the production of high-value crops. Focused on the development of automated systems through the integration of the Internet of Things (IoT) and machine learning, with the goal of increasing strawberry productivity and resource efficiency in controlled environments.

Professional Experience

Feb 2026–Current

Graduate Research Assistant

Department of Civil and Environmental Engineering
The Hong Kong University of Science and Technology
Hong Kong

Research focus: Utilizing nanobubbles in an IoT-
automated vertical farming system

PI: Dr. Samir Kumar Khanal

Feb 2025–Dec 2025

Undergraduate Research Intern

ICL Chemicals South America, São Paulo, Brazil

- Role: Conducting and analyzing experiments involving soil cultivation and hydroponics to test fertilizer efficiency

Oct 2023–Jan 2025

Exchange Student

The University of Minnesota

- Role: Assisting in the conduct of soil experiments and the processing of samples, with a focus on the efficient use of urea

Research Areas

- Hydroponic cultivation and Urban Agriculture
- Strawberry cultivation
- Nanobubble technology
- IoT-based Technology and Smart Farming
- Agriculture 4.0



Awards and Honors

1. **PostGraduate Studentship**–The Hong Kong University of Science and Technology – (Feb 2026- Current)

Certifications

1. Responsible Conduct of Research (1-RCR), The Hong Kong University of Science and Technology, Apr 2026
2. Lab Safety Training on Chemical, Biological and Fire Safety, Feb 2026

Relevant Courses

Graduate Courses

(MPhil in Civil and Environmental Engineering)

- Resource Recovery from Bioresources (CIVL4100P)
- Aquatic Chemistry (CIVL 5430)

Undergraduate Courses

(B.E. in Chemical Engineering)

- LSO0660 Soil Technology
- LSO0400 Soil Biology
- LSO0300 Soil Chemistry and Fertility
- LSO0310 Soil Physics
- LCB0103 Plant Morphology
- LCB0311 Plant Physiology
- LCB1402 Plant Ecology
- LGN0114 Cell Biology
- LCB1204 Systematic Botany
- LPV0448 Fruit Growing
- LPV0480 Vegetable Growing, Floriculture and Landscaping
- LES0208 Marketing